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CE F231 Fluid Mechanics

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October 21, 2022



Lecture 20: Fluid Kinematics

Vortex Flow

innovate

- Fluid mass moving along a circular path
- Naturally occurring or artificially generated
- Two types of vortex
 - Forced vortex
 - External torque is used to rotate the liquid
 - Free vortex
 - No external torque
 - Due to asymmetrical geometrical or physical boundary conditions
 - The fluid rotates due to some rotation previously imparted to it
 - Draining sink, bathtub, etc.
 - Whirlpool in rivers



Vortex Flow



<https://www.youtube.com/watch?v=eeQGKcmEX-Q>

<https://www.youtube.com/watch?v=lfZCnFwd-G8>

Vortex Flow (Cylindrical)

$u_r = 0 \rightarrow$ cylindrical; $u_r \neq 0 \rightarrow$ spiral

$u_r +ve$ (outward) $\rightarrow$ spiral out

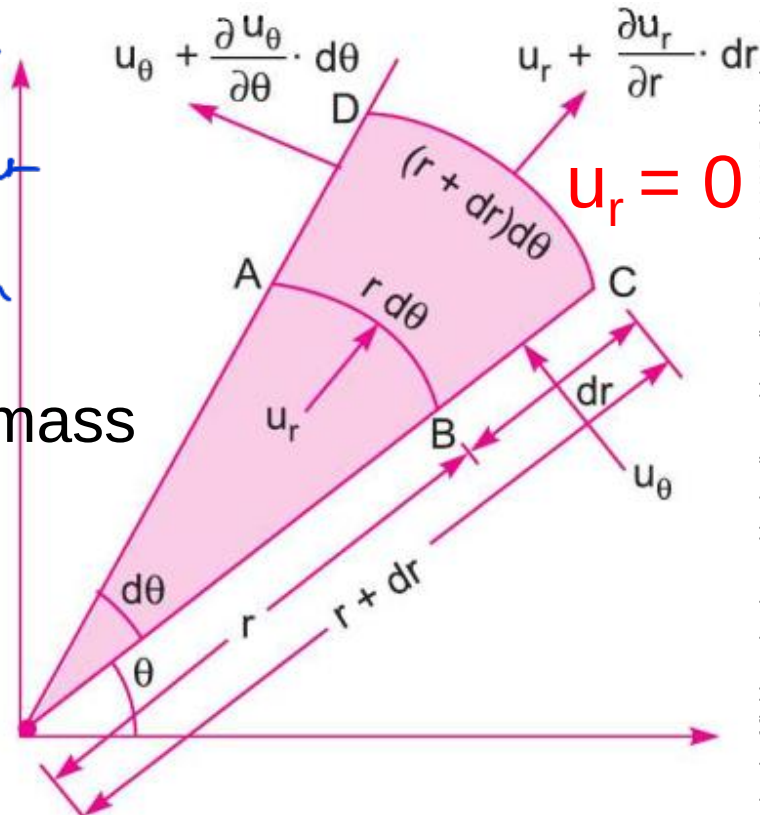
$u_r -ve$ (inward) $\rightarrow$ spiral in

Forced vortex

- Tangential velocity $u_\theta = r\omega \rightarrow$ spiral in
- $\omega =$ constant angular vel of whole mass
- $u_\theta/r = \omega = \text{constant}$

Free vortex

- Momentum = mu_θ
- Moment of momentum = $mu_\theta r$
- Time rate of change of moment of momentum = torque = 0
- $d/dt (mu_\theta r) = 0 \Rightarrow mu_\theta r = \text{constant} \Rightarrow u_\theta r = \text{constant}$



Vortex Flow

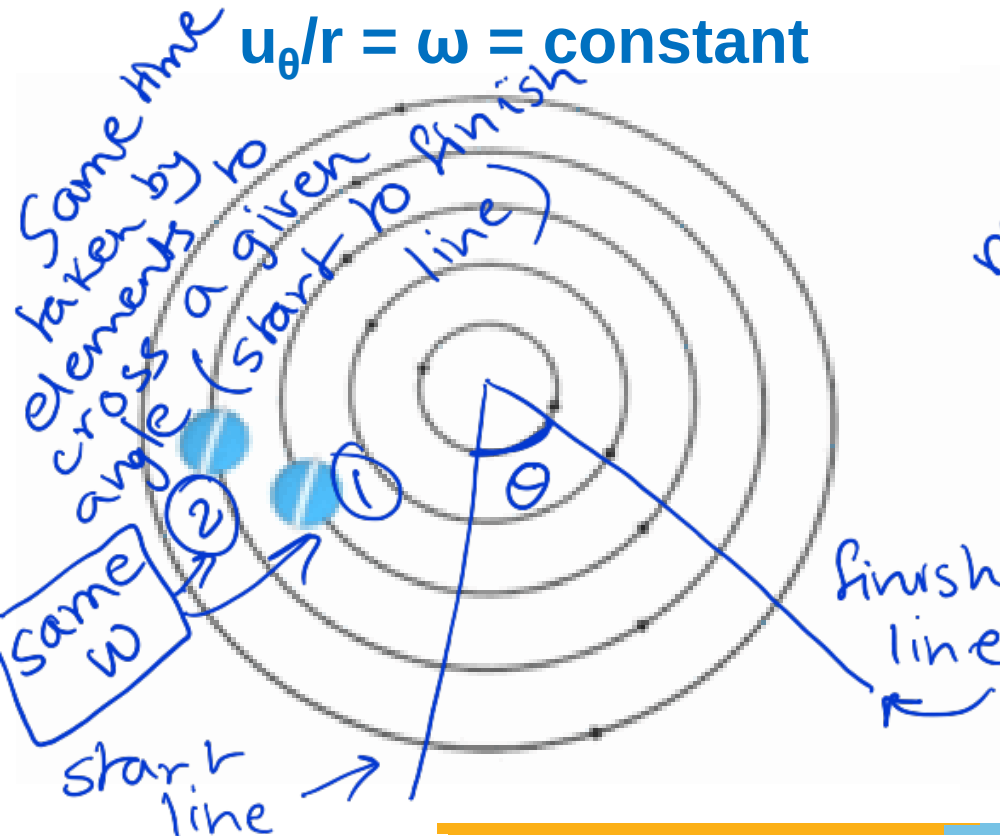
$$\frac{\theta}{t_1} = \frac{\theta}{t_2} \Rightarrow t_1 = t_2$$

Forced vortex

Rotational flow

Rigid body motion

$$u_\theta / r = \omega = \text{constant}$$



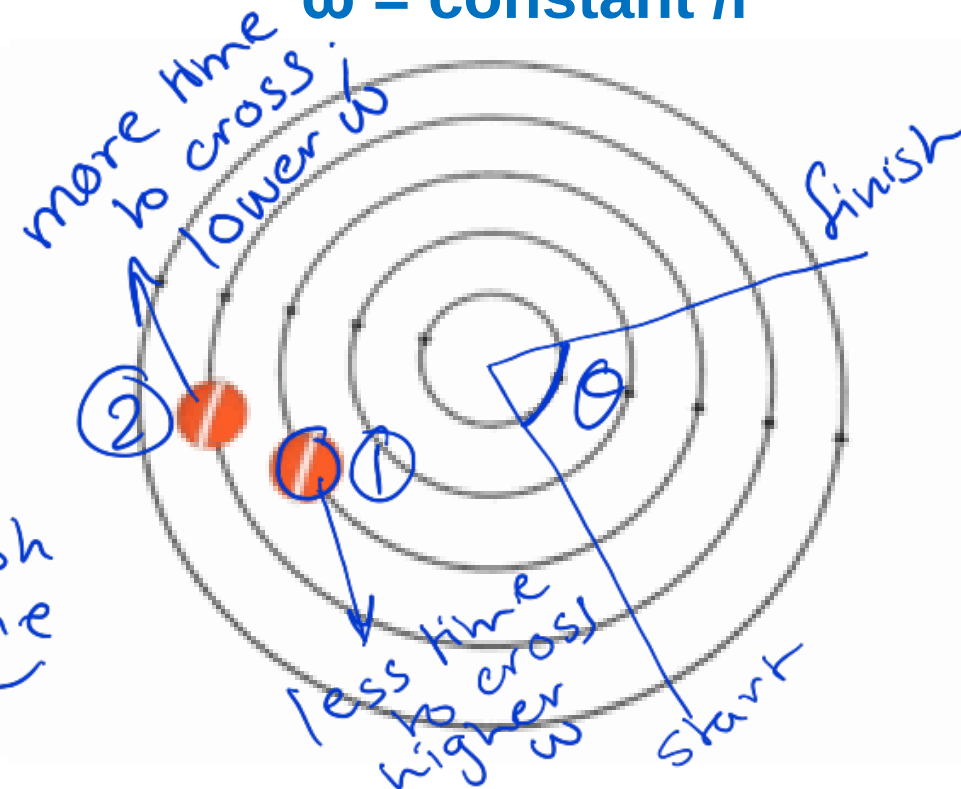
Free vortex

Irrotational flow

$$u_\theta r = \text{constant}$$

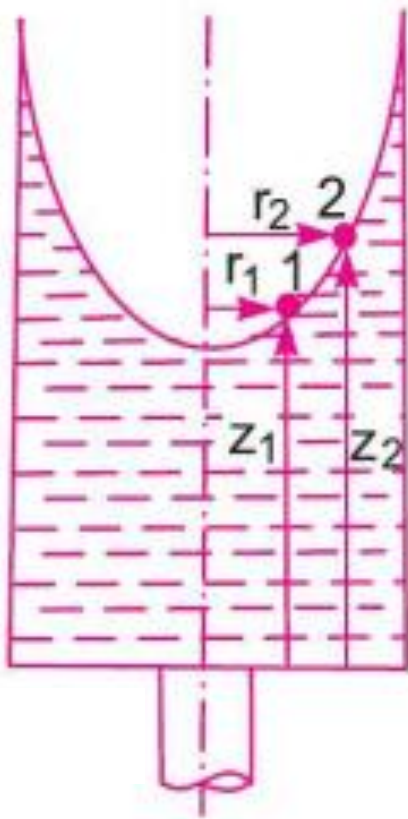
$$\omega = \text{constant} / r^2$$

$$\frac{\theta}{t_1} > \frac{\theta}{t_2} \Rightarrow \omega_1 > \omega_2$$

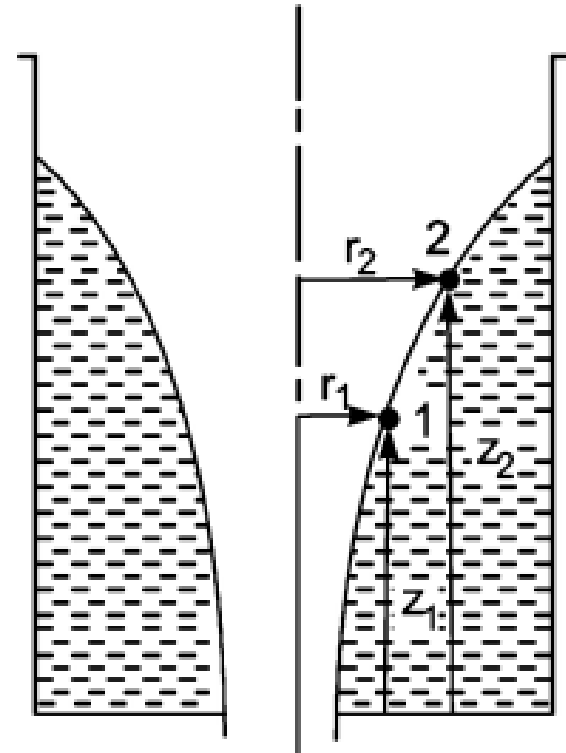


Vortex Flow

Forced vortex



Free vortex





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THANK YOU